

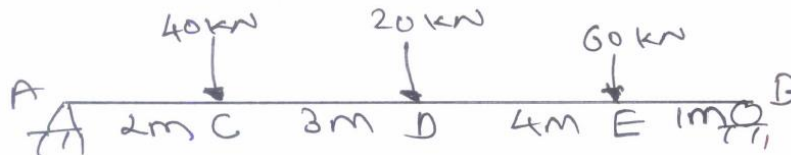
GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-IV (NEW) EXAMINATION – WINTER 2020****Subject Code:3140603****Date:11/02/2021****Subject Name:Structural Analysis-I****Time:02:30 PM TO 04:30 PM****Total Marks:56****Instructions:**

1. Attempt any FOUR questions out of EIGHT questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1** (a) Differentiate Plane frame and Grid **03**
 (b) State and explain principle of superposition. **04**
 (c) Using Conjugate beam method, find the slope and deflection in terms of EI at free end of the cantilever beam shown in figure. **07**



- Q.2** (a) Explain Maxwell's theorem of reciprocal deflections. **03**
 (b) Differentiate Conjugate beam and real beam **04**
 (c) Find slope at point A and B & deflection at point C in terms of EI for the beam shown in figure by Macaulay's method. **07**



- Q.3** (a) Differentiate between long and short column **03**
 (b) A masonry wall, 5 m high, is of solid rectangular section, 3 m wide and 1 m thick. A horizontal wind pressure of 1.2 kN/m^2 acts on the 3 m side. Find the maximum and minimum stress induced on the base, if unit weight of masonry is 22.4 kN/m^3 . **04**
 (c) A rectangular column section ABCD having $AB = CD = 400 \text{ mm}$ and $BC = AD = 300 \text{ mm}$ carries a compressive load 300 kN at corner B. Find the stress at each corner A, B, C and D and draw stress –distribution diagram for each side. **07**
- Q.4** (a) Discuss Stability checks for a dam. **03**
 (b) A "T" section is having flange with 100 mm and total depth 80 mm. The thickness of flange and web is 10 mm. The length of column is 3 m and it is hinged at both ends. If $E = 2.1 \times 10^5 \text{ N/mm}^2$, find Euler's buckling load. **04**
 (c) The external and internal diameter of a hollow cast iron column are 200 mm and 150 mm respectively. If column is hinged at both ends having a length of 4 m, determine the crippling load using Rankine formula. Take $f_s = 550 \text{ N/mm}^2$ and $\alpha = 1/1600$. **07**

- Q.5** (a) Explain advantages of three hinged arch over beam. **03**

- (b) Derive Euler's formula of critical load for column having both ends hinged **04**
 (c) A three hinged parabolic arch hinged at the support and at the crown has a span of 20 m and a central rise of 5m. It carries a concentrated load of 120 kN at 15 m from left support and a uniformly distributed load 20 kN/m over left half portion. Determine the moment, thrust and radial shear at a section 4 m from left support. **07**

Q.6 (a) Define Core of the Section. Derive and locate the same for a Circular cross section **03**

(b) A thin cylindrical shell of internal diameter d , wall thickness t and length L , is subjected to internal pressure p . Derive the expression for change in volume of the cylinder **04**

(c) A thin cylindrical shell of 600 mm diameter is 1500 mm long and 10 mm thick. It is subjected to internal pressure of 2 MPa. Calculate the change diameter, length and volume. Take $E = 200$ GPa and poisson's ratio = 0.27. **07**

Q.7 (a) State basic difference between fixed and simply supported beams. **03**

(b) A fixed beam of 10 m span carries central point load of 100 kN. Find fixed end moment equation using area moment method. **04**

(c) Using method of consistent deformation, analyze the propped cantilever beam shown in Figure, and draw shear force and bending moment diagrams. **07**



Q.8 (a) Define resilience, proof resilience and modulus of resilience. **03**

(b) Derive formula for strain energy due to sudden loading. **04**

(c) A steel bar 1 m length is subjected to a pull such that the maximum stress is equal to 150 N/mm^2 . Its cross section is 200 mm^2 over length of 950 mm and for the middle 50 mm length the sectional area is 100 mm^2 . If $E = 2 \times 10^5 \text{ N/mm}^2$, Calculate strain energy stored in the bar. **07**
